

Mehdi Zafari

Ph.D. Candidate, UC Irvine

Signal processing and AI/ML for next-generation (5G/6G) wireless systems

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Updated October 2026

Education



Ph.D. in Electrical Engineering and Computer Science

June 2024 – December 2028, expected

University of California, Irvine, CA, USA

GPA: 4/4

Advisor: Prof. A. Lee Swindlehurst



M.S. in Electrical and Computer Engineering

August 2021 – May 2024

Rice University, Houston, TX, USA

GPA: 3.92/4

Thesis: Distributed Multi-User Beamforming in the Presence of CFO: An Analytical and Experimental Study



B.S. in Electrical and Computer Engineering

September 2015 – September 2019

University of Tehran, Tehran, Iran

GPA: 3.7/4

Thesis: Design and Hardware Implementation of a MIMO-OFDM System Using High-Level Synthesis Techniques

Research interests

Wireless

communications

Cell-Free MIMO, Integrated Sensing and Communication, Resource Allocation, Beamforming

ML and optimization

AI/ML for Wireless, Distributed Optimization, Generative AI for Sensing and Wildfire Prediction

Experience

Jul 2024 – present

Graduate Student Researcher

EECS, UC Irvine

- Conduct research on reinforcement learning and graph neural network (GNN)-based models for cooperative ISAC networks
- Design distributed algorithms based on ADMM/consensus for beamforming and resource allocation in cell-free ISAC systems
- Develop large-scale network simulations and implement optimization/learning pipelines using Python/PyTorch, CVXPY, Gurobi
- Supervisor: Prof. A. Lee Swindlehurst

Jun – Sep 2026

Wireless AI Engineering Intern

Tiami Networks, Sacramento, CA

- Investigated ISAC channel models and constructed high-fidelity RF digital twins using NVIDIA Sionna and ray-tracing engines
- Analyzed spatial propagation characteristics to optimize sensing algorithms and facilitate integration with Open RAN architectures

Aug 2021 – Apr 2024	Graduate Research Assistant	ECE, Rice University
	- Conducted research on distributed multi-user MIMO beamforming for next-generation wireless networks - Performed over-the-air experiments using the RENEW massive MIMO testbed to evaluate distributed multi-user beamforming - Published four research papers and released over 200 datasets collected via the RENEW platform - Datasets: Distributed Multi-User MIMO Datasets on IEEE DataPort - Supervisor: Dr. Rahman Doost-Mohammady	
Jan – Apr 2024	Research Fellow	Rice D2K Lab
	- Mentored five undergrad students on a Rice EMS-sponsored project to analyze operational data across multiple domains - Developed time-series prediction models to anticipate future requirements in each area	
Dec 2018 – Dec 2020	Undergraduate Research Assistant	Array Processing Lab, University of Tehran
	- Conducted research on the hardware implementation of MIMO OFDM systems using the Xilinx ZedBoard platform - Designed full-stack OFDM system, from theoretical modeling in MATLAB/Python to hardware implementation via Vivado-HLS - Developed OFDM and SC-FDE systems on Xilinx FPGAs using High-Level Synthesis (HLS) - Supervisor: Dr. Aliazam Abbasfar	
May – Nov 2020	Part-Time R&D Engineer	Transportation Innovation Center, Tehran
	- Designed and implemented hybrid radio links (FM analog and OFDM digital) for radio-controlled camera drones - Researched telemetry system implementation strategies for vehicular communication in drone networks	
May – Nov 2020	Part-Time R&D Engineer	Sadr Electronic Dena, Tehran
	- Developed a 2.4GHz ISM band IoT system to enable remote control functionality for elevator central control boards - Engineered embedded C/C++ firmware for AVR and ARM microcontrollers with serial data transmission protocols	
Jul – Oct 2018	Full-Time Intern	Pionaria, Tehran
	- Simulated parabolic and patch antennas in HFSS and CST, improving VSWR and gain - Implemented point-to-point and point-to-multipoint wireless links using wireless network standards and software-defined radio (SDR) concepts	

Publications

Journal and magazine papers

CORDIS: A Scalable Coordinated Resource Allocation Framework for Distributed Cell-Free ISAC

M. Zafari, B. Ottersten, and A. L. Swindlehurst

IEEE TWC Submitted to IEEE Transactions on Wireless Communications

Under review

Robust Wildfire Forecasting under Partial Observability: From Reconstruction to Prediction

C. Yang, **M. Zafari**, Z. Duan, and A. L. Swindlehurst

IEEE TGRS

Submitted to IEEE Transactions on Geoscience and Remote Sensing

Under review

Multi-Domain Optimization Framework for ISAC: From Electromagnetic Shaping to Network Cooperation

R. Liu, M. Li, **M. Zafari**, B. Ottersten, and A. L. Swindlehurst

IEEE WCM

IEEE Wireless Communications (magazine)

An Analytical and Experimental Study of Distributed Uplink Beamforming in the Presence of Carrier Frequency Offsets

M. Zafari, D. Pandey, and R. Doost-Mohammady

IEEE TVT

IEEE Transactions on Vehicular Technology

Conference papers

TiamiTwin: A Digital Twin for Bistatic ISAC Drone Sensing, Validated Against Measurements

M. Zafari, S. Enayati, A. L. Swindlehurst, and A. Mukherjee

MILCOM 2026 Workshop

IEEE MILCOM 2026 Workshop on Integrated Sensing and Communication for Critical Infrastructure Protection (ISAC4CIP)

Accepted

ASSENT: Learning-Based Association Optimization for Distributed Cell-Free ISAC

M. Zafari and A. L. Swindlehurst

ICC 2026

IEEE International Conference on Communications (ICC)

Coordinated Decentralized Resource Optimization for Cell-Free ISAC Systems

M. Zafari, R. Liu, and A. L. Swindlehurst

Asilomar 2025

59th Asilomar Conference on Signals, Systems, and Computers



Best Paper Award Finalist

ADMM for Downlink Beamforming in Cell-Free Massive MIMO Systems

M. Zafari, D. Pandey, R. Doost-Mohammady, and C. A. Uribe

Asilomar 2024

58th Asilomar Conference on Signals, Systems, and Computers

ML-Based Feedback-Free Adaptive MCS Selection for Massive Multi-User MIMO

Q. An, **M. Zafari**, C. Dick, S. Segarra, A. Sabharwal, and R. Doost-Mohammady

Asilomar 2023

57th Asilomar Conference on Signals, Systems, and Computers

Robustness of Distributed Multi-User Beamforming: An Experimental Evaluation

R. Doost-Mohammady, **M. Zafari**, and A. Sabharwal

SAM 2022

IEEE 12th Sensor Array and Multichannel Signal Processing Workshop (SAM)

Datasets and software

Deep-Gen-Project: Spatiotemporal Generative Models for Wildfire Spread Prediction

M. Zafari, E. Finkelstein, and C. Yang

GitHub

GitHub repository

ASSENT-CellFree-ISAC: Simulation Framework

M. Zafari

GitHub repository

Distributed Cell-Free ISAC Simulation Framework

M. Zafari

GitHub repository

Distributed Multi-User MIMO Datasets

M. Zafari

IEEE DataPort

Talks

Mar 2026 Competition talk

When Wireless Signals Become Our Eyes

UCI Grad Slam finals, Irvine, CA

3rd place.

Oct 2025 Lecture

Coordinated Decentralized Resource Optimization for Cell-Free ISAC Systems

59th Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA

Best Paper Award finalist; also presented a poster in the Student Paper Contest.

Oct 2024 Lecture

ADMM for Downlink Beamforming in Cell-Free Massive MIMO Systems

58th Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA

Oct 2023 Poster

ML-Based Feedback-Free Adaptive MCS Selection for Massive Multi-User MIMO

57th Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA

Sep 2023 Poster

Distributed optimization for beamforming in massive MIMO systems

Texas Colloquium on Distributed Learning (TL;DR 2023), Rice University, Houston, TX

Jan 2023 Project presentation

DNN-based feedback-free adaptive modulation for massive MIMO

POWDER-RENEW Mobile and Wireless Week, University of Utah, Salt Lake City, UT

1st prize for the project.

Honors and awards

2026	Wireless History Foundation Scholarship Award	Wireless History Foundation
2026	Pia Gratton Endowed Graduate Student Award	UC Irvine
2026	1st Place, UCI EECS Annual Technology Showcase	UC Irvine
2026	3rd Place, UCI Grad Slam Competition	UC Irvine

2025	Best Paper Award Finalist, 59th Asilomar Conference	IEEE Asilomar Conference
2024	Graduate Dean's Recruitment Fellowship Award	UC Irvine
2022, 2023	SIAWE Scholarship Award for Distinguished International Students Society of Iranian-American Women for Education (SIAWE).	Rice University
2023	1st Prize, POWDER-RENEW Mobile and Wireless Week	University of Utah
2021	Rice University Graduate Fellowship	Rice University
2019	Exceptional Talent Award Top 5% GPA.	University of Tehran
2016	National Elites Foundation Fellowship	Iranian National Elites Foundation
2015	Ranked 47th, Iranian National University Entrance Exam (Konkour)	
2013	Ranked 3rd, Iranian National Mathematics Olympiad	

Service

2024–2026	Peer reviewer - IEEE Transactions on Wireless Communications (TWC) - IEEE Transactions on Vehicular Technology (TVT) - IEEE Transactions on Geoscience and Remote Sensing (TGRS) - IEEE Transactions on Aerospace and Electronic Systems (TAES) - IEEE Wireless Communications Letters (WCL) - IEEE Signal Processing Letters (SPL) - IEEE Communications Magazine
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Teaching

Fall 2026	Introduction to Control Systems + Lab EECS 160A/LA	UC Irvine
Spring 2025	Introduction to Digital Logic Laboratory	UC Irvine
Spring 2024	Applied Machine Learning & Data Science	Rice University
Fall 2023	Modern Communication: Theory and Practice	Rice University
Fall 2022	Modern Communication: Theory and Practice	Rice University
Spring 2020	Wireless Multimedia Communications	University of Tehran
Fall 2019	Signals and Systems Analysis	University of Tehran
Spring 2019	Digital Communications	University of Tehran

Teaching assistant for every course listed.

Skills

Key skills	Wireless Communications, Signal Processing, Optimization, Applied Machine Learning
Programming	C++, Python, TensorFlow, PyTorch, CVXPY, Gurobi, MATLAB, Verilog
Software-defined radios	USRP B210/X410, NVIDIA Aerial, GNU Radio, Iris SDR, Agora

Simulation tools NVIDIA Sionna, Ray-Tracing Engines, ns-3, Wireshark, Ansys HFSS, Simulink

Embedded and FPGA ARM STM32, DSP, Atmel AVR, Xilinx ZedBoard, Vivado HLS

Relevant coursework

Fall 2025	Deep Generative Models Grade: A	UC Irvine
Fall 2024	Machine Learning for Signal Processing Grade: A+	UC Irvine
Spring 2024	Optimization: Algorithms, Complexity, & Approximations Grade: A	Rice University
Spring 2023	Distributed Methods for Optimization and Machine Learning Grade: A+	Rice University
Fall 2022	Machine Learning (Introduction to Deep Learning) Grade: A	Rice University
Spring 2022	Advanced Wireless Networks Grade: A	Rice University
Fall 2021	Modern Communication: Theory and Practice Grade: A	Rice University
Fall 2017	Advanced Object-Oriented Programming Grade: A	University of Tehran

Selected conferences and workshops

Oct 2025, 2024, 2023	Asilomar Conference on Signals, Systems, and Computers	Pacific Grove, CA
Oct 2025	IEEE Military Communications Conference (MILCOM)	Los Angeles, CA
Sep 2023	Texas Colloquium on Distributed Learning	Houston, TX